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Creative Engineering Design

Professor: Kyu-jin Cho, Howon Lee

Date: December 1st, 2023

2023 Robot Project Final Report

I. Design intention

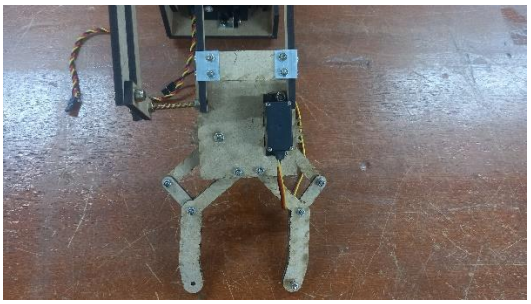
i) Harvest Robot

a) Wheels

The wheels of the harvest robot must first be able to climb the ramp well. In addition, it should be able to move freely and stably in a narrow stadium. Based on this purpose, our wheels were manufactured using two round PVC pipes on the front and two small, purchased wheels on the back, intending to move fast and stably in a stadium while being able to climb the ramp.

b) Gripper

Hence, the object is spherical our gripper is intended to grip object steadily with 2 stack MDF plate. The linkage of the gripper makes it easy to determine the angle of the gripper because it is a four-bar linkage with 1:1 ratio. It doesn't have advantage on torque however, servo HS255 had enough torque to handle both objects given in the task.

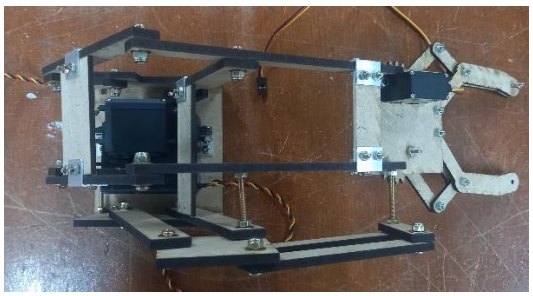


c) Linkage & Arm

For the arm of the robot that connects the gripper and allows the robot to move objects to the transport robot, out team decide to make a two DOF robot arm that resembles a crane. There are two servo motors (HS-815BB, HS-645MG) used in the arm part and each motor is connected to a 4bar linkage. The first motor (HS-815BB) lets the robot to lift the arm and that is the reason our team decided to use the strongest motor from the material since it carries the highest torque. Then, our second motor is connected to the second 4 bar linkage that controls the angle

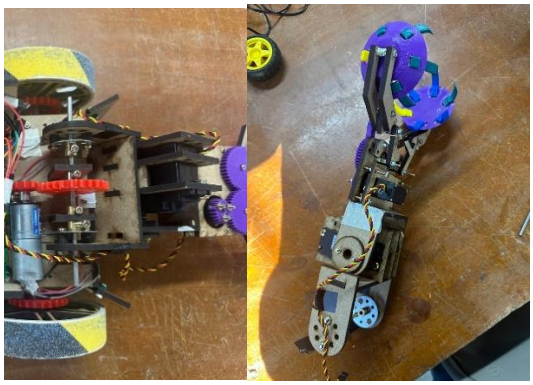
and height of the gripper. This allows us to have mor precise control over the arm. Also as seen in the picture, unlike other robot arms HS-645 is placed with HS-815 which makes the first motor to handle less torque. For the effective radius of the arm and the linkage, it will be further discussed in the kinetic analysis part (Part 3).





For our second design we intended have more robust arm. So, in our cad we had to case servo more firmly. It delivers the torque directly to the arm so, it was easier to track the torque applied to each servo. However, our second design exceeded the maximum torque of the 815-servo due to lack of current applied from the battery. Also, having one DC motor and three servo motors just for the arm was very much less power efficient. In order to fix this problem, we came up with our final arm.

Our final arm has one servo motor with one DC motor which has two degrees of freedom. Our DOF decreased from 3 to 2. However, our team thought that this degree of freedom is achievable with compact wheels. So, our final model has shorter arms which reduced momentum applied to the servo with no z axis rotation.



ii) Transportation Robot

The most important part of the scoring may be gripping fruits and put them in refrigerators. Since in last Line tracer Homework the car had limit speed so that they can catch white line signal for turning, we can't make our transport robot faster so that it can't transport more fruits. And making transport robot to go and return has risk, which means transport robot cannot trace the line well in an attempt, so all systems go wrong.

We designed our transport robot to be transformed near the refrigerators so that the harvest robot can put fruits near the room-temperature storages into the refrigerators using transformed transport robot. Also, since black sapphire is about 280g-which is far heavier than normal fruits, we decided not

to transport them. To put black sapphire in refrigerators with no transporting, we had to transform our transport robot.

But we still have to move some fruits using transport robot because each fruit that are put in the robot get 0.5 bonus point. So we also added the mechanism to transport fruits and put them in refrigerator when the transporter is unfolded.

II. Design alternatives and design iterations

i) Harvest robot

a) Wheels

The wheels were modified through a total of five processes as follows.

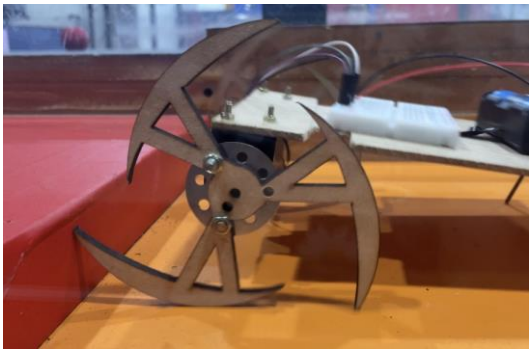
The first design alternative of the wheel is known as a “delta-wheel design.” It was the most visible type of wheel among the 'stair climbing wheels' on the Internet. This wheel is a wheel specialized for climbing stairs, in which three small wheels surrounding one center serve as one big wheel. When this meets the stairs, the front small wheel runs into a wall of the stairs, and the upper small wheel rotates over the stairs, so the upper wheel rolls over the stairs while the front wheel rolls under the stairs, leading it to climb the stairs. However, the wheel of this shape could only be made with gears, and it could not climb the stairs without strong driving from behind, so it failed.

(figure: delta-wheel design)



The second design alternative of the wheel was the “spoke tri-wheel design.” Three sharp curved wheel spokes protrude from the center of the wheel, with the upper wheel supporting the upper stairs and the lower wheel supporting the lower stairs, leading it to climb the whole stairs. It goes up the stairs with a mechanism very similar to the first design, but it is also difficult to go up the stairs without strong driving from behind, and the wheel in the form of a sharp spoke failed to support the heavy body of the robot and eventually broke. In conclusion, it failed.

(figure: spoke tri-wheel design)



The third design alternative of the wheel was the “large gear with rubber tubes.” This

wheel is a large gear-shaped wheel with a diameter of about 11cm, and it is a wheel that increases friction with the stairs by attaching a rubber tube(sliced) between each tooth using super glues. There were 24 teeth, and a tooth with a size that fit perfectly with the rubber tube (7mm outer diameter) was used. Unlike the previous two designs, it was a stable circular wheel, so it operated well on flat land. But it did not climb the ramp (which was also the case with the previous two laps), nor did it climb the stairs. In other words, it was a failed design.

(figure: large gear with rubber tubes)

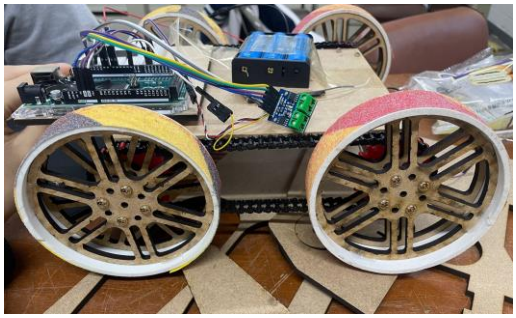


The following points were found through the previous three types of design. First, to stably drive the heavy body of the robot, the wheel must have a circular shape as a whole. There are many special types of wheels in the world to climb stairs, but it was difficult to move quickly and stably on flat surfaces other than stairs with such wheels because the number of materials and motors was limited. Second, to climb the stairs, the drive of the motor should not be transmitted only to one of the front and rear wheels. Many experiments have shown that it is more advantageous to connect the motor to the rear wheel than to connect the motor to the front wheel. It was confirmed that if the motor was connected to the rear wheel, the robot would climb the slope. However, to climb the stairs, the rear wheel first had to drive to make the front wheel to get over the stairs, and then the front wheel had to drive to make the rear wheel to get over the stairs, so both had to be driven. This led us to connect gears to the front and rear wheels, respectively, and to chain the two gears so that the rear wheels can roll together when the front wheels roll. Third, the performance depends on the surface material of the wheel. We have attempted to increase the frictional force using a rubber tube. However, in conclusion, we found that friction was needed more strongly, and accordingly, a rough sandpaper was attached to the wheel surface.

So the fourth alternative design of the wheel was the “chained-PVC round pipes.” Two pipes were used for the front wheel, and two for the rear wheel. Each pair of wheels arranged on the left and right were connected using gear and chains. Connection using teeth and chains is a strategy to make the motion of the rear wheel (driven using a DC motor) delivered to the front wheel, making it more advantageous when climbing a slope

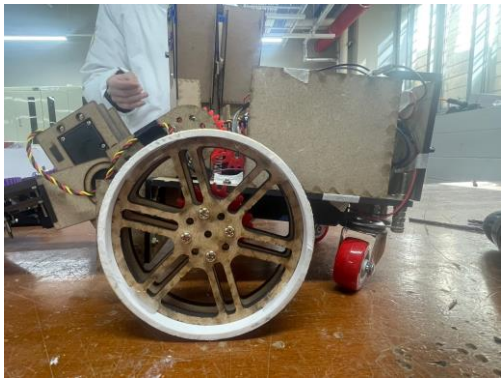
or stairs.

(figure: chained-PVC round pipes)



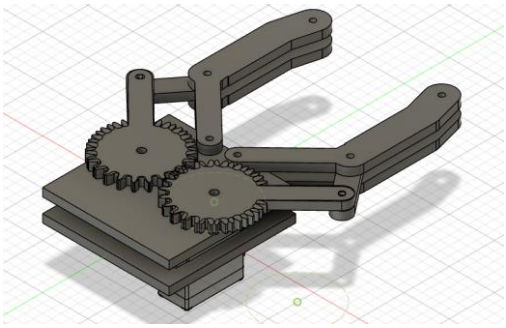
We used the 4th method of the wheel up to the 2nd evaluation, but we thought the harvest robot lacked maneuverability, so we made a major change. Rather than the traditional way of rolling on four big wheels, we put two big wheels in front of the robot and very small wheels behind them, which only serve to support the center of gravity. The reason for this change is to increase the mobility of the robot, as we thought it was important to reduce the time the robot moves in the actual game. We experienced less friction and were easier to control when rotating with two large wheels than when turning with four large wheels. However, it cannot climb the stairs. In usual situation, the two big wheels go to the front, however, when climbing the ramp, the two big wheels go to the back.

(figure: two big wheels and two small wheels)



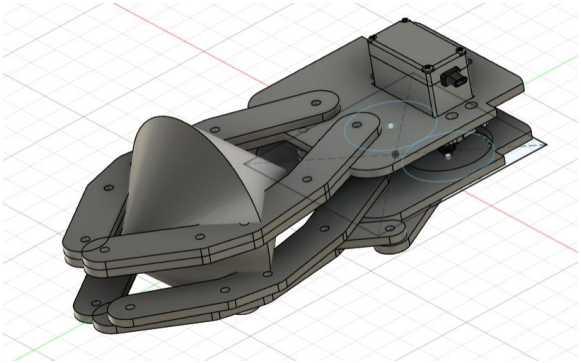
b) Gripper

At first, we wanted to make gripper fit the objects, so we added one more degree of freedom on the gripper. But it prevented proper linkage movement, so that it made a small normal force. After considering these problems, we modified the final gripper as the figure.



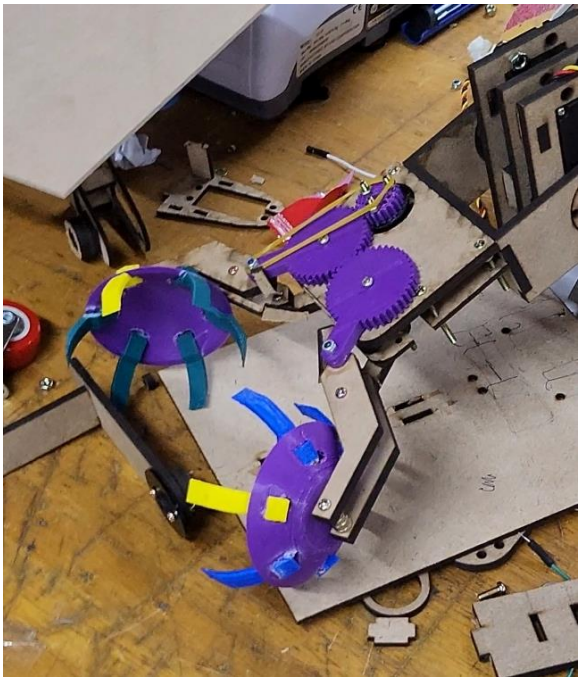
In the figure, there are two clamps attached to one side to allow the black sapphire to cut in between and receive additional force. Also, the tip of the gripper was slightly bent so that the sphere was easily held, and the design was made as simple as possible to minimize the weight.

And to compensate for the thin thickness in the middle, the size was increased, and the number of tongs was changed to eight.





However, we were so obsessed with minimization that we discovered a phenomenon that no matter how hard it tried to catch it, it lacked power. And the thickness problem or frictional force problem of the gripper is still not solved, so the shape of the gripper was made like a fly hell using 3D print. It is a shape in which small spines are alternately attached to a round circle in which a sphere or sapphire is to be placed. In addition, core parts such as gears were also made with 3D prints for robustness. This is because if we make the gear mdf, it continues to be damaged. As a result, we completed the design like the picture.



c) Arduino, Algorithm(I+II)

Gripper, robot arm control

The robot arm and gripper were controlled by individually controlling each motor using the channels of the RC transmitter. Due to the structural limitation of the robot arm, it only had vertical movement, and to easily correspond the arm's movement with the transmitter's controls, three servo motors were connected to channels 2, 5, and 6 of the RC transmitters (corresponding to the up/down motion of the right lever, left upper dial, and right upper dial of the RC transmitter, respectively). In the absence of signal input to the RC transmitter, the default value for each channel was approximately 1500. Therefore, when the input value to each channel was above 1520, the motor rotated clockwise, and when it was below

1480, the motor rotated counterclockwise, enabling control of the motor movement.

Body movement

The main body connects two DC motors connected to the wheels to channels 3 and 4 of RC Transmitter (up/down and left/right of RC Transmitter's left lever, respectively) so that when the left lever is moved up/down, it can rotate forward and backward, and left and right in place when it is moved left/right. The overall method is the same as when moving the robot arm, but in the case of body movement, due to the nature of the competition circuit, the wheel must move quickly and strongly in the first narrow task of lifting the black sapphire barrel, and after that, both fast and detailed movements are required. Therefore, the speed of the motor was determined by writing an equation with the channel transmission value as a variable to allow slow movement when the lever is moved a little and fast movement when the lever is moved a lot.

code:

```
if(ch[3]<1480){ //L하

analogWrite(LCCW, 0);

analogWrite(LCW, (1500-ch[3])/vvvv);

analogWrite(RCW, (1500-ch[3])/vvvv);

analogWrite(RCCW, 0);

}

else if(ch[3]>1520){ //L상

analogWrite(LCW, 0);

analogWrite(LCCW, (ch[3]-1500)/vvvv);

analogWrite(RCCW, (ch[3]-1500)/vvvv);

analogWrite(RCW, 0);

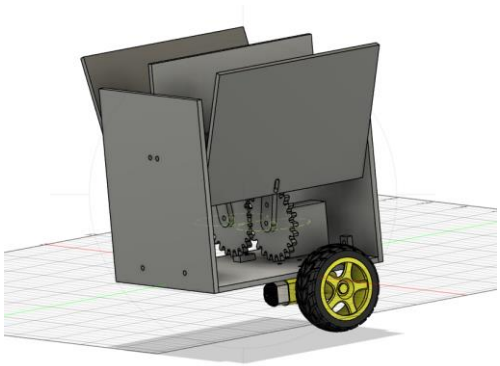
}
```

Arduino Circuit

We configured the circuit by parallelly connecting one battery pack to the Arduino board and two DC motors responsible for wheel movement. The remaining battery pack was parallelly connected to two servo motors and three DC motors for the robot arm section.

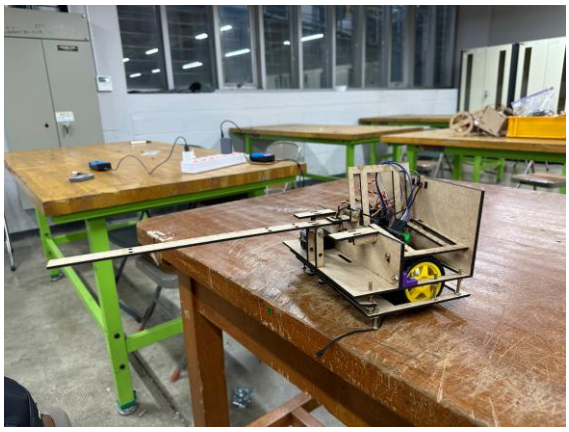
ii) Transportation robot

We thought about two available designs. The first one was just moving the transport robot and put fruits in refrigerators and put the remaining fruits in room-temperature storages.



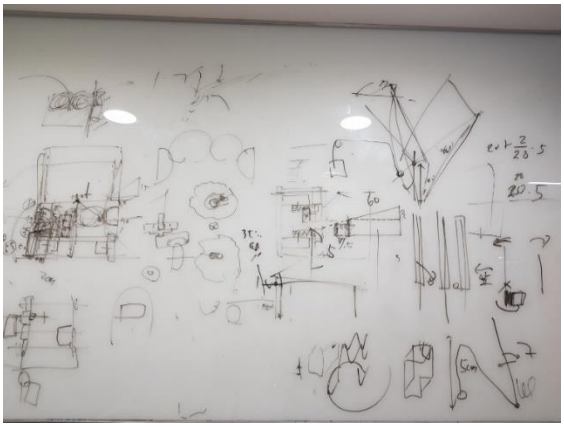


It worked well-they can easily put fruits in refrigerator. But before test 1, there were some teams that designed their transport robots that can put all the fruits in the refrigerators. So we decided to change our robot that can put fruits in the refrigerators, too. As a result, the second design of the transport robot was designed.

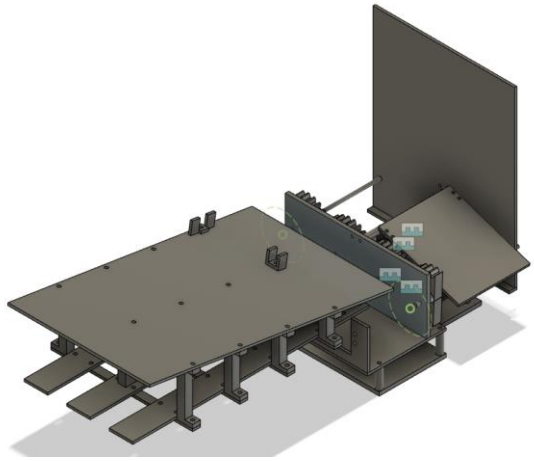


The second design of transport robot was simple. It could make rail bridge from the room-temperature storages. So, the harvest robot could put the fruits on rail, which make fruits to fall into their refrigerators. We called this ‘factory-like task’.

We next developed the second design more. There were four version of design in second car. The first design of the car was like the upper picture. It was designed to fall the bridge using hinge and then automatically slide the side raid so that the fruit hit the back of the transport robot and fall into each refrigerator. The hinge was good, the counterweight pully worked well, but the slider had so many friction that it couldn’t unfold the side rail. So we decided to change side rail to fall like the rail bridge using hinge-like mechanism.

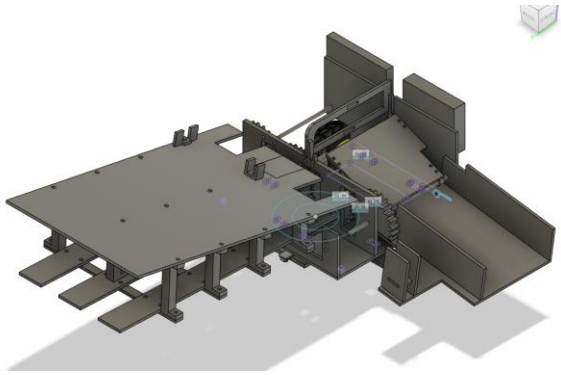


We discussed a lot how to change the transport robot. We wanted the transporter that can transport the fruits to get extra point, and also do the ‘factory-like task’. Also, when we stop the front rail bridge to flip using servo, we thought that there will be huge torque applied on the servo. Since there were some critical power issues on the harvesting robot in some teams, we changed that mechanism: from stopping the rail bridge to stopping the side bridge using gear. So, we made the car like below.



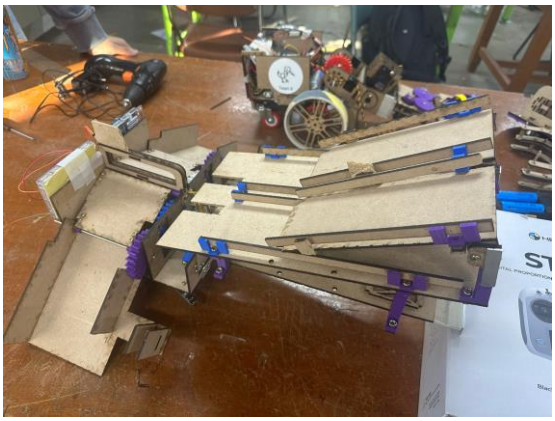
There are one inverse ‘V’ like shape at the back of the car, and we can put the fruit in there(there is no side bridge in that picture, but have one. You can understand when you watch the version 3 of the car). At the bottom of inverse ‘V’ shape lies the circuits of the linetracer.

That version worked well, but it had lots of difficulties modifying the circuits since in is under the inverse ‘V’ shape. We had to unfast the bolt that fix the shape to change the circuits. So we made the third version of transport robot.



The third version of the transport robot has shorter inverse ‘V’ shape compared to the second version. By shortening the shape, we made space to place the circuits at the back of the transport robot so that we can easily check the circuits if it’s okay.

It worked well, all of them are fine, but we had the last problem. The black sapphire was so heavy, and they get more friction comparted to other fruit because of the weight and the shape, they didn’t slip down the front rail bridge. So we made more slope at the highest part of the bridge. Then our final design came out.



III. Kinematic analysis and force analysis

i) Harvest Robot

a) Wheels

For the force analysis, when the front wheel meets the stairs and rolls, it receives a force acting upward by the upper stairs and a force acting in the direction of progress by the lower stairs, eventually being able to climb the stairs. However, in this situation, if the rear wheel is not driven, it cannot climb the

stairs. The reason for this is estimated as follows: The front wheel must contact the upper step while climbing the stairs, but without the drive of the rear wheel, the contact time is too short. In other words, because the robot (and the wheels) constantly shakes and moves, it does not stick tightly to the upper stairs and continues to shake without exchanging strength. Now after the front wheel climbs the stairs, the body of the robot moves forward again, and the same situation is placed on the rear wheel as the front wheel did. The rear wheel also climbs the stairs with the same principle, and similarly, it isn't easy to climb without the front wheel's drive. On all contact surfaces, the wheels may slip or roll. However, if the wheel's movement is mainly slipping when climbing the stairs, it only happens to swing in front of the stairs, so sandpaper tape was used to increase the frictional force between the contact surface and the wheel.

When moving on flat ground, the wheels are subjected to friction. In particular, when the rotation, in reality, there is no perfect rotation in place, so the wheel receives friction not only in a direction perpendicular to the axis but also in a horizontal direction. To reduce this friction, we finally reduced the number of large wheels with sandpaper from four to two.

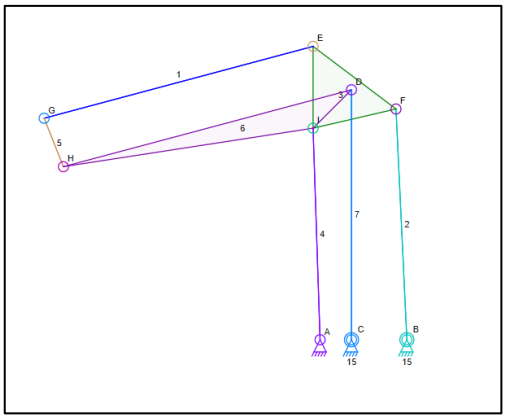
For the kinematic analysis, the two big wheels change the rotation motion to the translation motion. However, in reality, the sliding motion would be added. The sandpaper in the big wheel would reduce this. Also, the two small wheels can rotate 360 degrees (a direction perpendicular to the ground) so it rolls in the direction where the robot is moving.

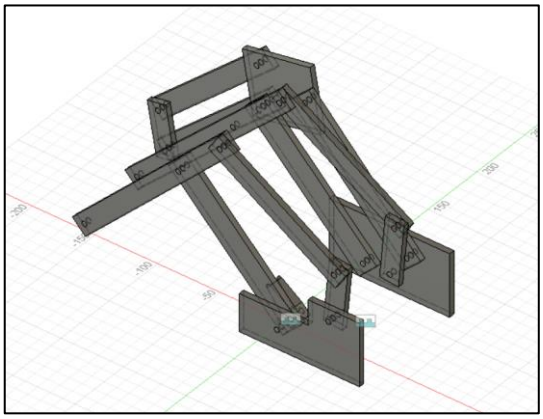
b) Linkage arm, Gripper

As seen in the figure below there are two four-bar linkages in the crane mechanism. The first linkage is connected to B compromised with bar 2,3 and 4. Maximum torque of servo HS-815BB is 19.8kg*cm at 4.8 volts and expected torque that is expected to apply on the motor is about 15.0kg*cm which is in the boundary of safety factor of 1.2.

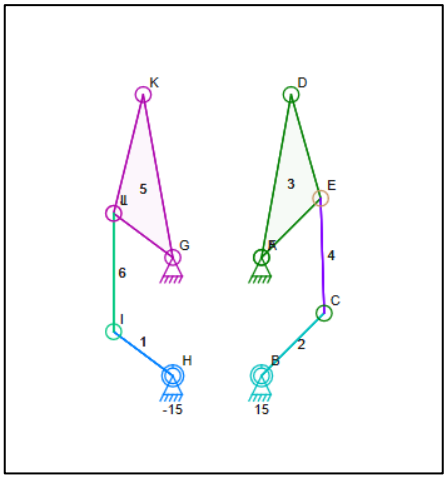
The second motor (HS-645MG) is linked to C which forms a four-bar linkage with 4,6 and 7.

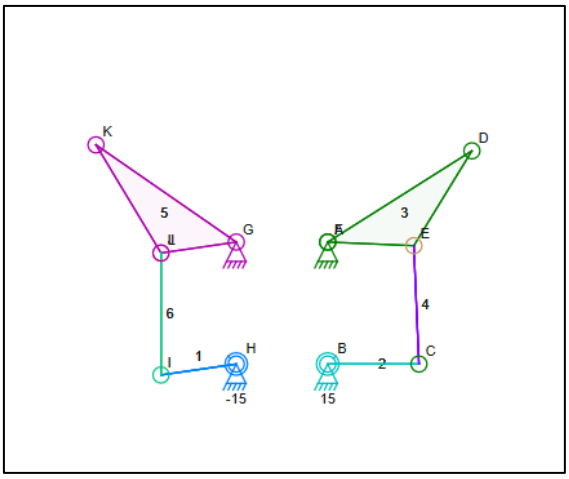
Limitation of the motor is set to 7.7kg*cm at 4.8 volts and expected torque is about 5kg*cm. This is due to torque advantage from the linkage. As seen in the cad model below there is a relatively short bar connected to the side plate. This gives 3times the torque advantage to the motor but, it limits the angle to the one third.





Linkage of the gripper is designed to deliver the same torque from the motor.



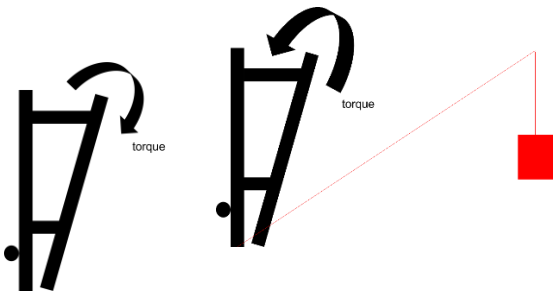


For linkage of our second arm, we intended to deliver the torque directly to the linkage. In order to do so, we made a socket of the servo and

ii) Transportation Robot

To make the rail bridge folded and unfolded, we made hinge at the bottom of bridge. This can make circular motion of rail bridge.

The first mechanism was as follow: The rail bridge is first fixed by servo motor and then released, unfolded, and make bridge. To do this, we had to use pulley to make the bridge unfolded smoothly.

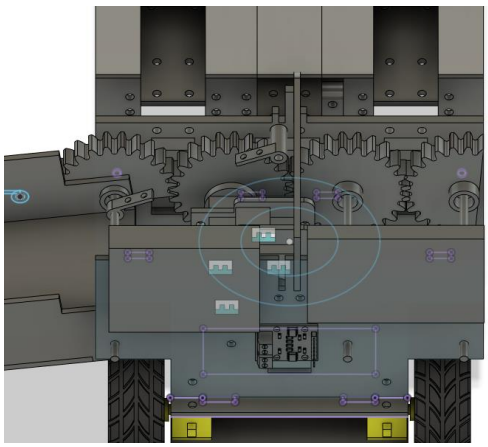


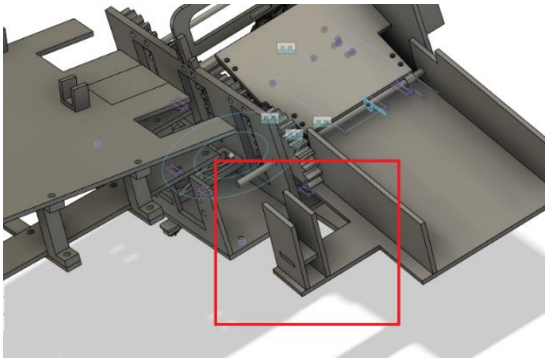
As you can see in the left figure, since the rail has more weight the torque of the bridge

is clockwise, which means the bridge isn't unfolded. By using pully like in the right figure, we can make torque counterclockwise, so the bridge is unfolded as we intended. And the torque given by pully is slightly bigger than the rail's torque. So the bridge can smoothly unfolded.

And using linkage, the stopper that make slider don't move get out, the slider slip and make plate for fruits to get in the refrigerators.

After some discussions, we change the mechanism of the unfolding. The first mechanism was to fix the rail bridge by servo, but we change that to fix the side bridge. The side bridge of the car is fixed by the servo using gear like this. The side gears which are connected to the side bridge are half-gear, which means they are free-unfolded when they are unfolded enough to free fall by torque of the gravity. This can make servo get less torque because the torque gets bigger and bigger when the gear rotates. Also, we had to consider the shape of the gear to be unfolded symmetrically.





That part of the side bridge grip the rail bridge not to be unfolded. When the side bridge is unfolded, the gripping part release the rail bridge and start to fall down.

Since rail bridge is very huge: about 500~600g, it still doesn't fall automatically when the gripping part is released. So, we had to use counterweights. But the first mechanism which used pully was very wasteful, since it makes the transport robot heavier, and it takes some space. So we change that counterweight to rubber band like picture below.



The rubber band is released when the bridge is unfolded and be tightened when the bridge is fixed by the gripping part. Using the rubber band, we can automatically unfold all

bridges using only one servo.

IV. Manufacturing difficulties and corrections

i) Harvest Robot

a) Wheels

The biggest manufacturing difficulty in wheels was, first, making wheels that could stably support heavy bodies, and second, making wheels that can be strongly fixed to the rotating shaft to perform stable rotation. First, in the case of supporting a heavy body, a single MDF plate with a 4.5mm thickness was not enough, so the method of connecting MDF plates with multiple layers of the same shape was used in the manufacturing process. In addition, the most up-to-date form is the use of PVC round pipe, which has also been confirmed to be stable. Next, the reason why the wheel should be strongly fixed to the rotating shaft and require stable rotation is that the robot can perform the movement as controlled by the user more accurate. So, the design was carried out with the engineering purpose of strongly fixing the wheel to the motor. By fixing the aluminum angle to the vehicle body, the wheel was fixed to the motor using this angle. (The motor is connected to the big wheels)

b) Gripper

The difficulty in making the gripper was that it was attached to the end of the arm, so it should be made as light as possible, but robust and strong. And in the case of general grippers, how to effectively widen the gap between the tongs on one side was also a difficult problem. Therefore, we used the smallest motor and tried to make the size of the gripper small, but the results failed repeatedly. Therefore, it was judged that it would be difficult to lift objects if the same stance was maintained, and it was possible to solve this problem using a 3D printer and a large motor. However, there is a problem that the parts printed with the 3D printer do not fit each other due to its tolerance. Since it was difficult to change the design and print it again, we solved this problem by widening the hole with a drill.

c) Linkage Arm

The hardest task manufacturing the arm was minimizing tolerance. We had limited 3d printing materials so, it was our task to make the linkage of the arm with MDF plates and fix them with nylon nuts. In order to make robust linkage to lift objects given in the task, each bar was made with at least two stacks of MDF plates and connected with aluminum brackets. Manufacturing L shaped plate was an obstacle for our team because it was our task to consider the thickness of the parts and smoothly rotate without overlapping each

other.

d) Arduino circuit

There were several issues:

Issue 1

There was a problem where the entire circuit would power off when the harvester moved vigorously or when excessive force was applied to the servo motors. It was determined that the motors had inconsistent grounding, and the battery pack might not be supplying sufficient power. To address this, the grounding was unified, and the problem was resolved by appropriately distributing two servo motors, three DC motors, and the Arduino board between the two battery packs.

Issue 2

A problem occurred where the DC driver, connected to two DC motors for movement, would consistently stop working during motion.

In response to this issue, we determined that the DC driver was struggling to control two DC motors simultaneously. To address this, we resolved the problem by using a total of three DC drivers and connecting only one DC motor to each DC driver.

ii) Transportation Robot

The most difficult part of manufacturing is limited resource, especially the MDFs. Since we can use only 2 400*800 4.5T MDF plates, and the most important part of the task is gripper and the arm, the 4.5T MDFs should be used there, not the transport robot. So, we had to design transport robot using 3T MDFs, which means we had to take more care of the structure stability. So, we made more supporting part on the bridge.

V. Result of each test & failure analysis & modifications

i) Harvest robot

a) Wheels

As mentioned above, there were a total of five design alternative models when divided largely, and the experimental results for each are as follows. Designs were modified through the trials.

First wheel (drive on the front wheels, adjusted height with screws instead of the rear wheel): the contact between the wheels and the floor is serrated, so it didn't roll well even on flat ground. Unable to climb stairs, failure to climb a slope. It is difficult for the three wheels to be fixed exactly in the center, so it has been observed that the wheels are slightly spinning in the air.

Second wheel (drive on the front wheels, adjusted height with screws instead of the rear wheel): Form made to climb stairs but unable to climb them. Determined to be due

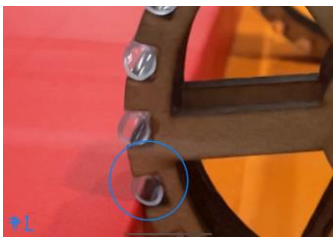
to no drive in the rear wheels. In addition, in the case of this form, we must determine the spacing or shape of the wheel bar (spokes) depending on the specific height of the stairs, which is judged to be inappropriate because the height of each step in the stadium is not constant. Also, during the experiment, the wheel bar broke while climbing the ramp. Determined to be unable to withstand the load on the body. Although it is possible to move on flat land, it is very shaken and cannot move accurately due to the nature of the wheel.

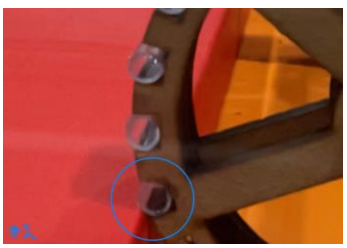
Third wheel (drive on the rear wheels, front wheels also installed): The friction of the rubber tube is not as great as we thought, showing the same result as without the rubber tube installed. It can climb the ramp (if we replace the drive with the front wheel, and in every other same condition it cannot climb). Most of the stairs cannot be climbed, but the lowest-height stairs can be climbed. Determined because the wheel is of a stable circular size and has sufficient size. Moving the flat is very stable. Also, in this trial run we could find out the motion of the wheels shaking and not having the exact contact with the stair steps (which was also mentioned in the force analysis.)

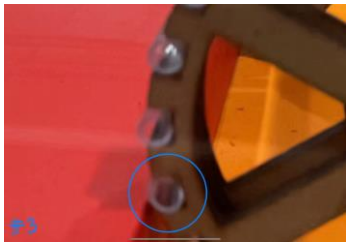
Fourth wheel (drive on the rear wheels, designed to allow the front wheel to move simultaneously as the rear wheel moves through chains and gears): It can climb ramps. It shows stable movement on flat land. Because sandpaper is attached to all wheels, there is a slight shaking when rotating in place. It can climb the lowest-height stairs reliably, and although it can't always climb the middle-height stairs, we confirmed that it can climb them if we add a load to the front of the harvest robot.

Fifth wheel (two big wheels and two small wheels): It can climb ramps (however when climbing ramps, the big wheel should be in the back, which means the robot should move backward). It shows a very stable movement on flat land. Almost no shaking when rotating in place. However, it cannot climb any stairs, and it shakes very much when going down the stairs.

(Figure: Shows the third wheel's trial run. It bounces off from the stair step, not having the exact contact, continually shaking. We can see that the wheel is backing off. The progress of time is in the order of figure 1,2, and 3)







b) Gripper

Our gripper picked the sphere-shaped fruit well, but the black sapphire did not. For this reason, we diagnosed the two main things. First, the weakest servo motor was used in the gripper. It would be nice to give a stronger force, but due to the nature of the gripper running at the end of the arm, we had to use a small motor to minimize the weight. Therefore, we had to find a way to catch it with a small servo motor. Second, the flat coefficient of friction of the mdf itself is not very high. For these reasons, objects were constantly missed in each test.

Afterwards, I felt that the gripper's frictional force was insufficient, so I attached sandpaper between them. However, the thickness of the gripper was still too thin to make a gripper that could only be held at a specific angle. To compensate for this, I put tissue between the sandpaper. The strong grip force was to make the gripper fit the object while holding the object. This certainly worked well when holding a spherical object, but the black sapphire could not pick it up, eventually changing the shape of the gripper once more.

In the last version, two modifications were made; first, a servo motor was used to increase the grip force, and gear ratio was used. And the tip of the gripper was made to fit in a circle using a 3D print, and a rod was attached to the side to make it stable. The result was a success, but as the force became stronger and the gripper were found to be stable, it was easy to lift not only the ball but also the black sapphire.

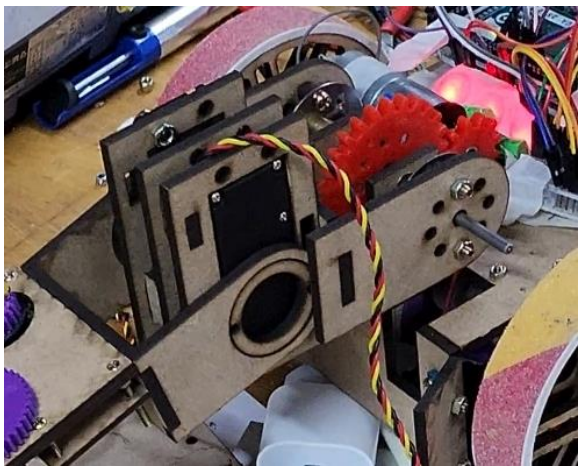
c) Arm

In the first test, the power of the Arduino was turned off every time the largest servo was moved. As a result of analyzing the cause of this phenomenon, it was confirmed that it was because not enough current was supplied to lift the linkage arm connected to the large servo. To improve this, we were able to solve this phenomenon by adding additional batteries provided

In addition, the linkage arm will be changed to the robot arm format instead of 4 bar linkage to maximize the range of operation. A DC motor was connected to the bottom to allow the arm itself to rotate 360 degrees, adding one degree of freedom. Through this, the so-called wheeled robot arm was completed.

. However, in the second test, the force increased as the arm lengthened, and the phenomenon of turning off when holding a black sapphire reappeared. Therefore, it was

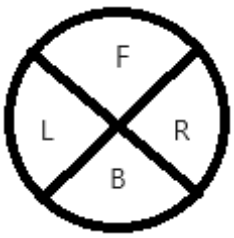
planned to reduce the strength by making the arms short, but if that happens, the advantage of the robot arm type will disappear due to too much loss of the range of operation. In the end, it was determined that the 815-servo motor was difficult to solve, so it was decided to use the 500:1 dc motor for the most powerful part. And in addition to the DC motor, the power was doubled by utilizing the 2:1 gear ratio. This could solve the power-off phenomenon. However, this required a complete revision. We had to design a whole new car without a few days left before the competition. It's as simple as possible, in a quick version. The mobility originally intended to be obtained from the arm was minimized by minimizing the body and changing the wheels to two-wheeled. At the same time, the arm's degree of freedom was reduced to two, and the overall shape is as shown in the following picture.



d) Code

An issue arose when the transmission values of channels 3 and 4 changed simultaneously, causing unintended movements due to conflicts between the forward/backward code and the left/right turn code. To address this, the movement was modified to halt when the transmission values of channels 3 and 4 changed simultaneously.

Difficulty arose in adjusting the movement so that when the lever is moved only slightly for slow chassis movement, only one of the transmission values of channels 3 and 4 could change. To resolve this, the system was adjusted to move forward and backward when the change in the value of channel 3 was greater than the change in the value of channel 4, and to turn left and right when the change in the value of channel 4 was greater than the change in the value of channel 3. In other words, the lever's movement space was divided to allow the chassis to move in distinct directions (F: Forward, B: Backward, L: Left, R: Right).



ii) Transportation Robot

In first evaluation, the transport robot can’t move since the IR sensors got wrong, and the

circuits burned that we could feel it hot when we turn it on. So, we have to change our IR sensors that are working well.

In second evaluation, the transport robot didn't work well. It didn't follow the line. After the match we checked our IR but it worked well. After few trials, we found out that the motor driver of the DC had poor-contact issue. We changed that connecting cable and the car worked well.

Also, after the first match the end of the bridge was nearly broken. We noticed that, and using glues and tape we fixed it. Also, we added some MDFs like picture to make that part harder.



Also, we found in trial that the car moves backward because of the momentum of the fruits falling down, especially for the black sapphire. So, we added the right angle frame at the front of the bridge so the bridge can be hung on the acrylic wall.

VI. What we have learned through this project

RKM: The CED was most impressive class that I've ever heard since elementary school. By taking this class, next time when manufacturing something I could have confidence doing things. This class enhanced my CAD and programming skills. Also, the most import thing were these: give and take / simple is best. I've learned that engineering is part of give and take, which means the perfect body which fill the full hexagon doesn't exist. I've also learned simple design and mechanism is best, most intuitive and easy to modify.

SHW: I think I learned two main things: one is the need for a leader and the other is that science and engineering are very different. It is often said that leadership is important, but in fact, I didn't know what it meant, but I learned that leadership is to ensure role distribution and lead team members toward goals when there is a big project. In that part, I felt that the person who diagnoses the team's problems and takes care of the mental state could be the leader. In addition, we thought that if we applied the physics formula with our hands or computers, the robot would be manufactured precisely, but we found that it was not fully implemented as we intended, starting with the robot's backlash. In the end, I found that an appropriate theory and trial and error should

be combined within a limited time.

KMJ: I realized that almost all the theoretical knowledge I knew, and the actual situation didn't fit together. It was a shock, but it was rather a fresh motivation, and I was able to enjoy the project thanks to it. The most important thing I realized is that rather than solving the problem, the most important thing is to figure out what the problem is. I found out that it takes a long time to figure out exactly where our robot has a problem, and that if we don't get the problem right, it takes longer to develop the robot. Also, I felt that I still lacked a lot of knowledge about physics, electrical circuitry, programming, etc., so I thought I should study hard. Also, I felt that communication with team members was very important, and I thought it was important to solve any task by communicating. It was a very precious experience, and I appreciate the TAs and professors.

SWS: I had a great experience working collaboratively with my team, holding meetings, and building a robot. It was a fulfilling journey where I utilized existing knowledge and filled in gaps with new information. Designing the robot, coding, and configuring the circuit directly provided significant academic achievements. Through numerous failures, I learned that, contrary to theory, there can be many variables in real-world scenarios. While solving problems was challenging, brainstorming solutions with team members and overcoming obstacles together proved to be highly rewarding.